

Tutorial

1. What should be the range values of the work function of a metal in order to be useful in a photo cell for detecting visible light?
2. An electron is accelerated through a potential difference of 400 V. Determine its de Broglie wave length.
3. The speed of an electron is found to be 1 km.s^{-1} within an accuracy of 0.02%. Calculate the uncertainty in its position
4. In a hydrogen atom, the electron in the $n = 2$ excited state remains there for 10^{-8} seconds on an average before making a transition to the ground state ($n = 1$).
 - (a) Calculate the uncertainty in energy of the excited state.
 - (b) What is the fraction of the transition energy?
 - (c) Compute the width of wave length corresponding to this.
5. Write down the normalized wavefunction if $\psi(x) = A \exp(-kx^2)$, where k and A are real constants over the entire domain.
6. Given $\psi(x) = A \sin(kx)$. Find the eigenvalues of the operator $\hat{O} = \frac{\partial^2}{\partial x^2}$. Find out whether $\hat{O} = \frac{\partial y}{\partial x}$ is an eigenoperator.
7. What are the time independent and time dependent Schrödinger's equations?
8. Which of the following is NOT a correct consequence of the Heisenberg uncertainty principle:
 - (A) The shorter the lifetime of an excited state of an atom, the less accurately can its energy be measured.
 - (B) An electron in an atom cannot be described by a well-defined orbit.
 - (C) The momentum of an electron cannot be measured exactly.
 - (D) Measurement of one variable in an atomic system can affect subsequent measurements of other variables.
 - (E) A harmonic oscillator possesses a zero-point energy.
9. Which of the following is NOT a correct aspect of the Born-Oppenheimer approximation:
 - (A) The electrons in a molecule move much faster than the nuclei.
 - (B) Excited electronic states have the same equilibrium internuclear distance as the ground electronic state.
 - (C) The electronic and vibrational motions of a molecule are approximately separable.
 - (D) Electronic energy curves serve as potential energy functions for nuclear vibrational motion.
 - (E) The typical amplitude of nuclear vibration is much smaller than that characterizing the motion of electrons.
10. Given a particle of mass m in a box of length L having the wave function $\Psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$, what is the energy of the level corresponding to $n = 4$?